

# Designing Reservation Reform Without Mass Unrest: An Agent-Based Model of Protest Against an Income-Only Quota in India

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github.com/utsavbalhara/reservation-reform-protest-model

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## Abstract

A recurring proposal in India is to replace caste-based reservation with a single income test at ₹8 lakh. We ask how much protest such a reform would provoke and which designs would keep it small. Three pieces of evidence replace assumptions of an earlier version of this analysis. First, under the actual creamy-layer and EWS rules the reform changes eligibility for about 6.8% of Indians who lose it and 3.1% who gain it. Second, an allocation model built on India's over-and-above choice rule and five years of JEE (Advanced) rank lists shows that a merged income pool would cost below-line SC candidates –72% of their IIT seats even though they keep eligibility. Third, an agent-based model on 640 Census districts is history-matched to machine-coded protest event counts from four episodes, including a null episode and a counter-mobilization.

The event data identify relative shock sizes far better than turnout: the 90% interval for the implied 2018 Bharat Bandh turnout is 7–539 lakh, and frozen out-of-sample predictions of deaths and of protest across states are weak. The level of protest the reform would provoke is therefore not identified, and moves by more than an order of magnitude across the assumptions the data leave open. The ranking of interventions is more stable. Keeping caste quotas with an income filter inside them changes peak turnout by –97%. It is the strongest single lever in all 11 structural variants and in every combination of the unidentified assumptions, in 98% of Morris design points, and with probability 95% under priors on how strongly each intervention acts. Compensating direct losers (–8%) and expanding seats on the EWS model (–9%) are the weakest levers; expansion does little because a merged pool still ranks SC and ST candidates against General candidates. Whether a consensus commission or grandfathering is the stronger end-state-preserving lever is not settled. Suppression raises deaths; whether it also broadens protest depends on an unidentified response to deaths.

**Keywords:** reservation policy; affirmative action; protest; agent-based modelling; history matching; event data; matching markets; India

## 1 Introduction

Few Indian policies generate as much contention as reservation. Quotas for Scheduled Castes (SC), Scheduled Tribes (ST) and Other Backward Classes (OBC) are defended as constitutional remedies for historical exclusion and criticized as caste-based rather than need-based. A proposal that surfaces repeatedly in public debate is to replace caste categories with a single income criterion. Its supporters present it as fairer; its opponents present it as the end of caste-based recognition.

This paper does not take a position on whether that reform is desirable. It asks two narrower questions that any government contemplating it would need answered:

- Q1.** How much street protest would an income-only reform provoke, and from whom?
- Q2.** Which choices of policy design, sequencing and political process would keep that protest small, and which commonly used responses would make it worse?

These questions are hard to answer from data alone. The reform has never been attempted, protest turnout in India is poorly measured, and the mechanisms involved interact non-linearly: loss aversion, group identity, contagion between neighbours, organization and repression. An agent-based model makes the assumed mechanisms explicit and lets alternative designs be compared under identical conditions (Epstein, 2002; Granovetter, 1978). It also makes it easy to mistake assumptions for findings. Much of this paper is therefore about separating the two: which inputs can be grounded in data, which conclusions follow from the dynamics rather than from how an intervention was coded, and which survive changes to the model's structure.

**Contributions.** To our knowledge, and after the searches documented in Appendix A, this is the first agent-based model of reservation politics and the first study to compare reservation reform designs by the protest they would provoke. Its mechanisms are not new; the claim is narrow. The paper contributes:

1. **An allocation accounting of the reform.** Eligibility is not the same as seats. Using the over-and-above choice rule that India uses to implement vertical reservation (Aygün and Turhan, 2023b; Sönmez and Yenmez, 2022), a latent merit model fitted to five years of JEE (Advanced) rank lists, and the actual eligibility rules for the OBC creamy layer and the EWS quota, we estimate how an income-only pool would redistribute IIT seats. The reform is not a change for “2.8% of the population”: 6.8% of Indians lose eligibility, 3.1% gain it, and below-line SC and ST candidates, who keep eligibility, lose most of their seats to below-line General candidates.
2. **A model calibrated to event data from several episodes.** A district-level synthetic population (640 Census districts) is history-matched (Andrianakis et al., 2015; Vernon et al., 2010) to machine-coded protest event counts (GDELT) for four reservation-related episodes, with a hand-audited relevance sample, reported deaths, and two observation-model anchors. Leave-one-episode-out predictions are frozen and hashed before being scored.
3. **A comparison of interventions that states what is input and what is emergent.** Each lever's effect size gets a prior; we report the probability that each lever is strongest, decompose levers into channels, sweep the unidentified ratio of symbolic to material grievance, and rerun the comparison across a structural ensemble and a Morris screening.
4. **A transparent development record**, including two specifications that were badly wrong and how a historical validation test caught them.

Section 2 sets out the legal architecture, the eligibility accounting and the mobilization record. Section 3 reviews related work and Section 4 the theory. Section 5 presents the allocation model, Section 6 the protest model, and Section 7 the episode data and calibration. Section 8 records the model's development. Sections 9–12 present the experiments, results and robustness analyses, and Sections 13–16 discuss implications, limitations, ethics and reproducibility.

## 2 Background

### 2.1 The legal architecture of reservation

The Constitution permits special provisions for socially and educationally backward classes and for SCs and STs (Articles 15(4) and 16(4)), and reserves legislative seats for SCs and STs (Articles 330 and 332). Central quotas stand at 15% for SC, 7.5% for ST and 27% for OBC (Deshpande, 2013; Galanter, 1984).

Income tests entered the system through the courts. In *Indra Sawhney v. Union of India* (1992), the Supreme Court upheld the 27% OBC quota introduced after the Mandal Commission, but

excluded the “creamy layer” of advanced OBC families and capped total reservation at 50%. The creamy-layer exclusion was reaffirmed for OBC admissions to central educational institutions in *Ashoka Kumar Thakur v. Union of India* (2008). The Constitution (One Hundred and Third Amendment) Act, 2019 added a 10% quota for Economically Weaker Sections (EWS) of the General category. The Supreme Court upheld it by 3:2 in *Janhit Abhiyan v. Union of India* (2022). To introduce EWS without reducing other categories, the government told the Court it had approved about 2.14 lakh additional seats in central institutions, roughly a 25% increase in intake (The Tribune, 2022).

The two income tests are not the same test, and neither is a simple cut at ₹8 lakh of family income.

- **OBC creamy layer.** The income-and-wealth test uses a ₹8 lakh threshold, but income from salaries and from agricultural land is not counted (DoPT, 1993). Separate status criteria exclude, for example, children of senior (Group A) officers. An OBC family whose gross income exceeds ₹8 lakh mainly through salary or farm income is therefore non-creamy and eligible today.
- **EWS.** Eligibility requires family income from all sources below ₹8 lakh and excludes families owning 5 acres or more of agricultural land, a residential flat of 1,000 square feet or more, or residential plots above set sizes (DoPT, 2019). Some General-category families below the income line are therefore ineligible today.

SC and ST quotas remain without any income test. In *State of Punjab v. Davinder Singh* (2024), a seven-judge bench permitted sub-classification within SC and ST quotas, and several judges suggested that a creamy-layer principle should also apply to them. The Union government stated it would not apply a creamy layer to SCs and STs. Census 2027 will enumerate caste for the first time since 1931 (PIB, 2025).

## 2.2 What an income-only test changes: eligibility

We take the reform to be one test on gross family income from all sources at ₹8 lakh, with no category and no asset test. Table 1 sets out who changes status. Two shares are not observed: the share of above-line OBC families who are non-creamy today (prior 0.3–0.7, central 0.5), and the share of below-line General families who fail the EWS asset tests (prior 0.05–0.25, central 0.15). With the central values, 6.8% of the population loses eligibility (range over the priors 5.2%–8.3%) and 3.1% gains it (1.0%–5.1%). An earlier version of this analysis, which treated both income tests as the same gross-income cut, reported 2.9%; that figure counted only SC and ST families above the line.

**Table 1:** Eligibility before and after a single gross-income test at ₹8 lakh (district-based synthetic population; central values of the two unobserved shares).

| Segment                                        | Share of India | Today                | After reform                  |
|------------------------------------------------|----------------|----------------------|-------------------------------|
| SC and ST families above ₹8 lakh               | 2.9%           | Reserved             | <b>Unreserved</b>             |
| OBC above ₹8 lakh, non-creamy today            | 3.9%           | Reserved             | <b>Unreserved</b>             |
| SC and ST families below ₹8 lakh               | 22.4%          | Reserved (own quota) | Reserved (merged pool)        |
| OBC families below ₹8 lakh                     | 38.0%          | Reserved (own quota) | Reserved (merged pool)        |
| General below ₹8 lakh, EWS-eligible            | 17.2%          | Reserved (EWS)       | Reserved (merged pool)        |
| General below ₹8 lakh, failing EWS asset tests | 3.1%           | Unreserved           | <b>Reserved (merged pool)</b> |
| OBC creamy layer and General above ₹8 lakh     | 12.6%          | Unreserved           | Unreserved                    |

## 2.3 What an income-only test changes: seats

Eligibility is only half of the material change. Today the reserved categories hold separate quotas. Under the reform the SC, ST, OBC and EWS quotas merge into one pool for all below-line families, so below-line SC and ST candidates compete with below-line General and OBC candidates for the same seats. Whether they gain or lose depends on how the pool is sized and on where each group sits in the merit distribution, which differs substantially by category. Evidence on caste and income as targeting criteria points the same way: caste-based targeting in engineering admissions reaches students poorer than the upper-caste applicants they displace (Bertrand et al., 2010), and caste remains a marker of disadvantage among poor households (Deshpande and Ramachandran, 2019a). Section 5 estimates the seat changes directly.

## 2.4 Reservation mobilization in India

Reservation has repeatedly produced large mobilizations. In 1990, implementation of the Mandal Commission's OBC quota provoked sustained student protests. In 2006, OBC quotas in central higher education provoked the Youth for Equality movement. Demands for backward-class status drove the Patidar agitation in Gujarat (2015), the Jat agitation in Haryana (2016), the Kapu agitation in Andhra Pradesh (2016), the Gujar agitation in Rajasthan (2019) and the Maratha Kranti Morcha in Maharashtra, whose largest march, in Mumbai on 9 August 2017, drew estimates of 3 to 5 lakh participants.

Four episodes concern SC and ST protections or the income test directly and are used to calibrate the model (Section 7):

- **2 April 2018.** In *Subhash Kashinath Mahajan v. State of Maharashtra* (March 2018), the Supreme Court diluted arrest provisions of the SC/ST (Prevention of Atrocities) Act. No one lost a quota. A Bharat Bandh followed on 2 April; 11 to 14 people died. It was organized with little central leadership or party backing: the first call came from a local leader in Punjab and spread through WhatsApp and local Dalit organizations (Daniyal, 2018). Same-day reports spoke of “thousands” of protesters (Al Jazeera, 2018); later commentary of “hundreds of thousands” (Shepherd, 2018). Parliament restored the provisions later that year.
- **6 September 2018.** Upper-caste organizations called a Bharat Bandh against the restoring amendment. It was strongest in Madhya Pradesh, Bihar and Uttar Pradesh, with no deaths reported.
- **21 August 2024.** After *Davinder Singh*, a Bharat Bandh demanded that no creamy layer or sub-classification apply to SC and ST quotas. Several opposition parties backed it, yet it drew a mixed response, and SC and ST communities that expected to gain from sub-classification opposed it.
- **January 2019.** The EWS amendment passed with little street protest, which makes it a useful null episode.

From these episodes we draw stylized facts that the model should reproduce or respect:

- S1.** A purely symbolic threat to SC/ST protections can produce national, bandh-scale mobilization (2018).
- S2.** Mobilization is organized around bandh calls and concentrated on those days.
- S3.** Large mobilizations do not require party backing (2018 was locally organized), and party backing does not guarantee them (2024). Party backing is at most one amplifier among several.
- S4.** Groups split when a reform creates winners inside them (2024).

**S5.** Deaths occur at some large mobilizations and not others, and states often concede afterwards (2018).

S3 revises an earlier version of this paper, which took opposition-party backing to accompany every large mobilization. The 2018 record does not support that, and the calibration below treats party backing as an episode-specific input.

### 3 Related work

**Reservation design.** The closest formal work is in matching theory. Sönmez and Yenmez (2022) formalize how India implements vertical and horizontal reservations, and Aygün and Turhan (2020, 2023a) study the de-reservation of unfilled OBC seats in IIT admissions; Aygün and Turhan (2023b) characterize the over-and-above rule axiomatically. We use their choice rules rather than inventing an allocation mechanism. Empirical work compares caste and income as targeting criteria (Bertrand et al., 2010; Deshpande and Ramachandran, 2019a,b). An incentivized experiment finds that SC and ST beneficiaries are judged less competent even when affirmative action is income-based (Deshpande et al., 2026), which bears on whether an income test would remove the symbolic charge of reservation. None of this work models the political response to reform.

**Protest models.** Threshold models (Granovetter, 1978; Schelling, 1978), critical-mass theory (Marwell and Oliver, 1993) and preference falsification (Kuran, 1991) explain why collective action can be all-or-nothing and hard to predict. Epstein (2002) began a line of agent-based models of civil violence; Srbljinović et al. (2003) model ethnic mobilization, and Lemos (2018) compares model output with the size, duration and recurrence of events in the Social Conflict Analysis Database. Thron and Jackson (2015) caution that such models may display generic self-organized criticality rather than the mechanisms their authors intend. That critique applies here, and it is one reason we test mechanisms against several episodes, report which results are inputs, and use a structural ensemble. On India specifically, Wilkinson (2004) shows that electoral incentives shape whether states prevent or permit communal violence, which motivates the concession and policing mechanisms.

**Validating agent-based models.** Windrum et al. (2007) survey empirical validation of agent-based models. We follow the history-matching approach of Vernon et al. (2010) and Andrianakis et al. (2015), which rules out parameter sets inconsistent with data rather than fitting a single best set, and which suits a stochastic model with several unidentified shock sizes. Simulation-based inference (Cranmer et al., 2020) is an alternative that we leave for future work. For global sensitivity analysis we use Morris screening (Campolongo et al., 2007; Morris, 1991; Saltelli et al., 2008). The model description follows the spirit of the ODD protocol (Grimm et al., 2020).

**Event data.** GDELT (Leetaru and Schrodt, 2013) codes events automatically from news reports. It over-reports events in capitals and media centres (Hammond and Weidmann, 2014), which our observation model allows for. ACLED (Raleigh et al., 2010) covers India from 2016 and would be a natural cross-check; we did not use it here.

## 4 Theory and hypotheses

### 4.1 Loss aversion and the politics of retrenchment

Prospect theory holds that losses loom larger than equivalent gains, with a loss-aversion coefficient near 2.25 (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992). Pierson (1994,

1996), building on Weaver (1986), argues that retrenchment is harder than expansion because costs are concentrated, immediate and traceable while benefits are diffuse. Governments that retrench successfully rely on blame-avoidance strategies: *obfuscation*, which lowers the visibility of cuts; *division*, including grandfathering, which spares current beneficiaries; *compensation* for losers; and *blame-sharing* through broad coalitions. We model the transparent versions of these strategies. Obfuscation is excluded on ethical grounds (Section 15).

## 4.2 Group identity and symbolic threat

Social identity theory holds that part of individuals' self-concept derives from group membership, and that perceived threats to a group's status provoke collective responses (Tajfel and Turner, 1979). For SC and ST communities, caste-based reservation is a constitutional acknowledgement of historical injustice as well as a material benefit. Removing caste as the criterion is therefore a symbolic threat independent of any individual's eligibility. Stylized fact S1 is direct evidence that this channel operates at national scale; the 2018 bandh followed a ruling that removed no one's quota.

## 4.3 Thresholds, cascades and organization

In threshold models, each individual joins once the share of others participating exceeds a personal threshold (Granovetter, 1978; Schelling, 1978). Aggregate outcomes can then be highly sensitive to the distribution of thresholds, and cascades can be global or fizzle depending on small changes (Watts, 2002). Critical-mass theory stresses that a small, highly motivated core can make collective action viable for everyone else (Marwell and Oliver, 1993), and preference falsification means that latent support can be invisible until a cascade begins (Kuran, 1991). Resource-mobilization and political-process theories stress that organizations and political opportunities convert latent grievance into action (McAdam, 1982; McCarthy and Zald, 1977; Tarrow, 1994).

## 4.4 Repression and backfire

The relationship between repression and dissent is contested: repression sometimes deters and sometimes escalates (Davenport, 2007; Lichbach, 1987). Hess and Martin (2006) describe conditions under which repression backfires by generating outrage. For India, Rydzak (2019) finds that internet shutdowns are associated with increases in violent collective action rather than decreases, and Rydzak et al. (2020) find little evidence that shutdowns quell protest elsewhere. Because the literature supports both deterrence and backfire, the model does not assume either: the response of turnout to deaths is a parameter whose prior runs from deterrence to strong backfire.

## 4.5 Hypotheses

- H1.** Grandfathering and a glide path reduce protest by deferring material loss (loss aversion).
- H2.** Expanding seats reduces protest, following the EWS 2019 precedent.
- H3.** Retaining caste categories with an income filter inside them reduces protest by reducing symbolic threat.
- H4.** Sub-classification favouring the most-deprived reduces protest by splitting the coalition (S4).
- H5.** A data-first commission with cross-party consensus reduces protest by reducing organization and symbolic threat.

- H6. Compensating direct losers reduces protest (Pierson’s compensation strategy).
- H7. Credible statutory guarantees reduce protest by limiting feared future changes.
- H8. Internet shutdowns reduce protest.
- H9. Heavy policing reduces protest.

## 5 The allocation model: who gains and loses seats

### 5.1 Choice rules

India implements vertical reservation with the *over-and-above* rule (Aygün and Turhan, 2023b; Sönmez and Yenmez, 2022). Open-category seats go first to the highest-merit applicants of any category; each reserved category’s seats then go to its highest-merit remaining members. Reserved-category candidates who would win an open seat on merit therefore do not use up a reserved seat. We apply the same rule to the reform, with the open share unchanged at 40.5% and one reserve category, “family income below ₹8 lakh”, holding the rest (59.5% in the main case; 49.5% and 69.5% as variants).

Actual IIT admission is a centralized multi-programme assignment in which top reserved-category candidates often take reserved seats at preferred programmes. For the aggregate number of seats by category, this makes the outcome closer to a *reserved-first* order, in which reserved seats are filled before open seats. We check both orders against the actual category-wise allotments (Section 5.4) and use the one that reproduces them as the main case. Horizontal reservations and candidates’ preferences over programmes are not modelled.

### 5.2 Latent merit

The JEE (Advanced) Common Rank List records each qualified candidate’s rank and category. We transcribed the lists for 2019–2025 (Appendix B) and model each category’s candidates as normally distributed on a common latent merit scale, with the General category fixed at mean 0 and standard deviation 1. The number of candidates per category comes from the official reports. Given the parameters, rank  $r$  sits at the point  $z$  where the expected number of candidates above  $z$  equals  $r - \frac{1}{2}$ , and the probability that the candidate at that rank belongs to category  $c$  is proportional to  $N_c \phi_c(z)$ . The means and standard deviations maximize the likelihood of the observed categories at every rank. The 2019 and 2020 lists failed an internal consistency check and are excluded; the fits use 2021–2025.

### 5.3 Family income

Family income is not observed per candidate. Each category’s share of candidates above the line is drawn from a prior: General 0.70–0.95 (the category includes OBC creamy-layer candidates and General candidates above the EWS line), OBC-NCL 0.10–0.40 (above the line on gross income but non-creamy), SC and ST 0.10–0.45; GEN-EWS candidates are below by definition. A merit gradient (0–0.8 on the logit scale per standard deviation of merit) lets richer candidates sit higher in each category’s distribution. A draw is kept only if the implied share of admitted IIT students above the line is 28–45%, the range consistent with the parental-income tables of the JEE reports once income growth is allowed for. We make 300 draws per year and keep 1,500 in total.

### 5.4 Validation

In sample, the fitted category shares match the observed shares by rank decile to within 3.5 percentage points. Out of sample, the GEN-EWS and OBC-NCL rank lists end at the same

aggregate-marks cutoff, as do the SC and ST lists, so the latent cutoffs implied by each list's length should agree within each pair. They agree to within 0.012 standard deviations for GEN-EWS and OBC-NCL, and to within 0.03 for SC and ST in every year except 2023 (0.126). Under the reserved-first order, the model's status-quo seat counts by category match the actual IIT allotments to within 406 seats per category and year; the open-first order misses by up to 2,386 seats. The reserved-first order is therefore the main case. NEET, where the same approach would cover medical admissions, is excluded because the fitted tails failed a check against the top-ranked candidates.

## 5.5 Results

Table 2 gives the seat changes by segment in the main case. The reform does not only take seats from SC and ST families above the line. Below-line SC candidates lose  $-72\%$  of their seats and below-line ST candidates  $-85\%$ . The gains go to below-line General candidates ( $+158\%$  for those who fail the EWS asset tests today and  $+71\%$  for GEN-EWS) and, modestly, to below-line OBC candidates ( $+23\%$ ). The reason is the merit distribution: at the same income, SC and ST candidates sit lower on the fitted merit scale, so a pool open to all below-line families admits relatively fewer of them. The direction of every change is the same under both processing orders and all three pool sizes (Appendix C).

**Table 2:** IIT seats by segment under today's quotas and under a merged income-only pool (reserved-first order, pool 59.5%). Medians over 1,500 draws of the income overlay across 2021–2025; brackets give 90% intervals. Relative material change is the change in admission chances per candidate in units of the loss of an above-line SC candidate (positive = loss); it is the material input of the grounded protest model.

| Segment                                    | Seat change | 90% interval     | Relative material change | 90% interval       |
|--------------------------------------------|-------------|------------------|--------------------------|--------------------|
| SC, above the line                         | $-96\%$     | $-98$ to $-93$   | 1.00                     | 1.00 to 1.00       |
| SC, below the line                         | $-72\%$     | $-75$ to $-68$   | 0.66                     | 0.50 to 0.75       |
| ST, above the line                         | $-98\%$     | $-99$ to $-96$   | 1.09                     | 1.00 to 1.17       |
| ST, below the line                         | $-85\%$     | $-88$ to $-81$   | 0.84                     | 0.62 to 0.96       |
| OBC (non-creamy), above the line           | $-87\%$     | $-92$ to $-79$   | 0.74                     | 0.64 to 0.84       |
| OBC, below the line                        | $+23\%$     | $+10$ to $+37$   | $-0.16$                  | $-0.27$ to $-0.08$ |
| General, EWS-eligible                      | $+71\%$     | $+56$ to $+87$   | $-0.45$                  | $-0.57$ to $-0.35$ |
| General, below the line, not EWS-eligible  | $+158\%$    | $+144$ to $+189$ | $-2.61$                  | $-3.13$ to $-1.84$ |
| General (incl. OBC creamy), above the line | $-10\%$     | $-16$ to $-7$    | 0.18                     | 0.11 to 0.30       |

The same model evaluates the two levers that are themselves allocation rules. Expanding seats by a quarter, the scale used to introduce EWS in 2019, softens the below-line SC loss to  $-63\%$  but does not remove it, because the merged pool still ranks SC candidates against General candidates. Keeping the caste quotas but closing them to above-line SC and ST families, whose seats go to below-line candidates of the same category, reverses the sign for the below-line majority: below-line SC candidates gain  $+32\%$  and below-line ST candidates  $+37\%$ . Brazil's 2012 quota law (Law No. 12.711) combines criteria in this way, with racial sub-quotas inside an income- and school-based quota.

## 6 The protest model

The model is described here in the order of the ODD protocol's overview and details (Grimm et al., 2020). It has two specifications. The *stylized* specification is the original model, retained

as a reference and for comparison with earlier results. The *grounded* specification replaces its assumed inputs wherever data allow; Table 3 lists the differences.

## 6.1 Population

The grounded population has  $N = 120,000$  agents, each representing about 12,200 people of a population of 146 crore. Agents are allocated to the 640 Census 2011 districts in proportion to population. Within each district the SC and ST shares are the district's own (Census 2011); the rest are split between OBC and General with the state's ratio of OBC to OBC-plus-other household heads in NFHS-5. National shares are then SC 16.7%, ST 8.6%, OBC 45.7% and General 29.0%. SC and ST agents are assigned with equal probability to a better-off tier or a most-deprived tier. The probability of being above the ₹8 lakh line is 30% for General and 17% for OBC; for SC it is 19% in the better-off tier and 5% in the most-deprived tier, and for ST 16% and 4%. These shares order groups consistently with survey evidence on the caste gradient of income and wealth (Bharti, 2018; PRICE, 2023). Above-line OBC agents are creamy layer with probability 0.5, and below-line General agents fail the EWS asset tests with probability 0.15.

Each agent draws an identity strength  $\iota_i \sim \text{Beta}(2, 3)$  and a standard-normal threshold score  $z_i$ . Neighbourhoods of about 100 agents are formed within state and group blocks ordered by district, so a neighbourhood spans about three districts. A share  $h$  of agents live in single-group neighbourhoods and the rest in mixed neighbourhoods whose composition matches the national mix;  $h$  is estimated in calibration. The stylized population has the same structure without districts, with  $h = 1$  (complete segregation), NFHS-4 group shares (OBC 42%, General 32.8%), and no distinction between the creamy layer and the income line.

## 6.2 Grievance

Agent  $i$  in eligibility segment  $k$  has a material change  $m_k$  (positive for a loss). In the grounded specification  $m_k$  is the relative material change of Table 2, which puts the loss of an above-line SC candidate at 1. In the stylized specification it is 1 for SC and ST above the line, 0.35 for SC and ST below, 0.25 for OBC below,  $-0.15$  for General below, and 0 otherwise. Losses are weighted by loss aversion,  $L(m) = \lambda m$  for  $m > 0$  and  $m$  otherwise. Symbolic threat  $s_g$  is 1.0 for SC, 0.9 for ST, 0.45 for OBC and  $-0.4$  for General, times 0.8 for the most-deprived tier; in the grounded specification it is scaled by the 2018 episode's estimated magnitude (Section 7). Grievance on day  $t$  is

$$G_{i,t} = w_m L(m_{k(i)}) + w_s \iota_i (\tilde{s}_i + \mu_{g,t} \mathbb{I}[\tilde{s}_i > 0]), \quad (1)$$

where  $\tilde{s}_i$  is the agent's tier-adjusted symbolic threat and  $\mu_{g,t}$  the accumulated martyr effect for the group.

## 6.3 Social influence, organization and decision

Let  $n_{c,t}$  be the share of neighbourhood  $c$  protesting on day  $t$  and  $v_{g,t}$  the national share of group  $g$ . Social pull saturates:

$$S_{i,t} = \beta_n \tanh\left(\frac{n_{c(i),t-1}}{n^*}\right) + \beta_v \tanh\left(\frac{v_{g,t-1}}{v^*}\right). \quad (2)$$

Organization contributes  $a o_g b_t$ , where  $o_g$  is group organizational capacity (SC 0.6, ST 0.4, OBC 0.4, General 0.1),  $a$  the opposition-party amplifier and  $b_t$  the mobilization level: 1.0 on bandh days and 0.15 otherwise. The agent protests with probability

$$P_{i,t} = \left[ 1 + \exp\left(-\frac{G_{i,t} + S_{i,t} + a o_g b_t - c_t - \theta_{i,t}}{\tau}\right) \right]^{-1}, \quad \theta_{i,0} = \bar{\theta} + \sigma_\theta z_i, \quad (3)$$

where  $c_t$  is a turnout cost imposed by heavy policing on bandh days. Each day of protest raises an agent's threshold by the fatigue increment  $f$ .

## 6.4 Bandh calls, concession and backlash

In the stylized specification bandhs are called on fixed days (3, 13 and 26). In the grounded specification, organizations call a bandh for the next day once the turnout that a bandh would draw, given today's grievance and social pull, reaches 10 lakh. The first call can take effect on day 3; after each bandh there is a refractory period of 10–13 days, and there are at most four calls. The government concedes with a daily hazard that rises logistically with pressure, defined as the largest daily turnout so far in crore plus deaths so far divided by 20 (midpoint 1, width 0.25), up to 15% a day. A concession cuts material loss and the threatened groups' symbolic threat by 60%, cancels any scheduled bandh, and gives the General category a symbolic threat of 0.3, which can produce a counter-mobilization.

## 6.5 Deaths and the martyr effect

In the stylized specification, deaths on day  $t$  are Poisson with mean  $\kappa$  times the day's protester-days in crore, and every death raises  $\mu_g$  by  $\delta = 0.012$  for each group in proportion to its share of protesters, up to 0.35. In the grounded specification deaths are negative binomial with dispersion estimated in calibration, the rate on bandh days is ten times the ordinary-day rate, and deaths are split into police-attributed deaths (half at baseline) and others. A response parameter  $\rho$  multiplies the martyr effect:  $\rho = 1$  is the reference backfire,  $\rho = 0$  no response, and  $\rho < 0$  deterrence. Heavy policing raises only police-attributed deaths.

## 6.6 Uncertainty and paired sampling

Each Monte Carlo run draws a plausible world. In the stylized specification this means uniform multipliers on  $\lambda$  (0.7–1.3), symbolic threat (0.8–1.2),  $a$  (0.85–1.15),  $\beta_n$  and  $\beta_v$  (0.8–1.2 each) and  $\kappa$  (0.5–2.0), plus  $\mathcal{N}(0, 0.1)$  on  $\bar{\theta}$ . In the grounded specification, the parameters informed by the episodes ( $\bar{\theta}$ ,  $\sigma_\theta$ , the reform's symbolic threat, the bandh-day death rate and dispersion) are instead taken from one parameter set drawn at random from those that survive calibration, and the reform's symbolic threat is additionally multiplied by 0.8–1.2. Run  $r$  uses world seed  $1000 + r$  and campaign seed  $5000 + r$  for every scenario, so scenario comparisons are paired. Effects are reported as the median over runs of the paired ratio (scenario over baseline in the same world), with 95% percentile-bootstrap intervals over runs (Efron and Tibshirani, 1993), and as the share of runs in which the scenario exceeds the baseline.

**Table 3:** Stylized and grounded specifications.

| Component                        | Stylized                                     | Grounded                                                          |
|----------------------------------|----------------------------------------------|-------------------------------------------------------------------|
| Population                       | National shares; single-group neighbourhoods | 640 districts (Census 2011, NFHS-5); mixing $h$ estimated         |
| Eligibility                      | One gross-income line for every test         | Creamy-layer and EWS asset rules                                  |
| Material change                  | Assumed per segment                          | IIT allocation model (Table 2)                                    |
| Symbolic threat                  | Assumed                                      | 2018 episode magnitude $\times R$ , $R = 1$                       |
| $\bar{\theta}$ , $\sigma_\theta$ | Tuned to 20 lakh–1 crore peak                | History-matched to four episodes                                  |
| Bandh calls                      | Fixed days                                   | Called when expected turnout is credible                          |
| Concession                       | None                                         | Pressure-dependent hazard; General backlash                       |
| Deaths                           | Poisson; all trigger martyr effect           | Negative binomial, split by attribution; bandh-day rate estimated |
| Random streams                   | One stream                                   | Separate streams for decisions, deaths and events                 |

**One campaign.** Build the population; draw a world; apply the scenario’s interventions. Then for  $t = 0, \dots, 39$ :

1. compute  $G_{i,t}$ ,  $S_{i,t}$  and  $P_{i,t}$  for every agent;
2. draw each agent’s protest decision and raise the thresholds of those who protest;
3. update neighbourhood and national turnout; decide whether a bandh is called for tomorrow and whether the government concedes;
4. draw deaths and update the martyr effect.

**Outputs:** peak-day protesters, unique participants, protester-days, deaths, bandh days called, concession day, and daily series by group and state.

## 7 Episode data and calibration

### 7.1 Event counts

Protest turnout in India is rarely counted, and the estimates that exist differ by an order of magnitude (for 2018, from “thousands” to “hundreds of thousands”). Instead of calibrating to a headcount, we calibrate to the number of protest events reported. From the GDELT 1.0 daily event files we take every event with CAMEO root code 14 (protest) located in India on each episode’s core days, and keep those whose source article URL matches episode-specific keywords (Appendix B). Keyword filters admit irrelevant events: a fuel-price bandh on 10 September 2018, unrelated state bandhs in August 2024, and a general strike during the EWS amendment. For each episode a sample of matched URLs was labelled as relevant or not from the article’s URL text; the posterior precision on core days,  $\text{Beta}(1 + \text{relevant}, 1 + \text{irrelevant})$ , multiplies the count. To absorb GDELT’s changing coverage, counts are expressed per 1,000 GDELT events located in India on the same days. Table 4 lists the episodes and targets.

**Table 4:** Episodes and calibration targets. Events: keyword-matched GDELT protest events on core days; precision from the audited sample; rate per 1,000 India events on the same days (median and 90% interval). Role: M mechanism episode (simulated), A anchor for the observation model, O held for description only.

| Episode                              | Role | Events | Precision | Rate per 1,000   | Deaths | Party backing |
|--------------------------------------|------|--------|-----------|------------------|--------|---------------|
| SC/ST Bharat Bandh, 2 April 2018     | M    | 308    | 0.96      | 31.3 (27.7–32.4) | 11–14  | 0.2           |
| Upper-caste bandh, 6 September 2018  | M    | 77     | 0.86      | 7.5 (6.1–8.3)    | 0      | 0.1           |
| SC/ST Bharat Bandh, 21 August 2024   | M    | 90     | 0.85      | 10.2 (8.2–11.4)  | 0      | 0.6           |
| EWS amendment, January 2019          | M    | 247    | 0.09      | 1.6 (0.3–4.2)    | 0      | 0.0           |
| Maratha march, Mumbai, 9 August 2017 | A    | 56     | 0.96      | 6.4 (5.6–6.6)    | 0      | 0.5           |
| Patidar agitation, August 2015       | A    | 77     | 0.86      | 2.5 (2.0–2.7)    | 10–11  | 0.1           |
| Jat agitation, February 2016         | O    | 204    | 0.96      | 5.5 (4.9–5.7)    | 30–31  | 0.2           |
| Kapu agitation, January 2016         | O    | 9      | 0.89      | 1.5 (1.0–1.7)    | 0      | 0.1           |
| Maratha quota protests, 2018         | O    | 214    | 0.96      | 8.9 (7.9–9.2)    | 2–8    | 0.4           |
| Gujjar agitation, February 2019      | O    | 54     | 0.86      | 1.1 (0.9–1.2)    | 0      | 0.1           |

## 7.2 Simulating an episode

Each mechanism episode is simulated as a two-day campaign, an announcement day and a bandh day, with no material change. Its shock is a symbolic threat vector (which groups feel threatened, and in which direction) scaled by an unknown magnitude. For 2018 the vector is (1, 0.9, 0, -0.2) for SC, ST, OBC and General; for the September 2018 upper-caste bandh (-0.3, -0.3, 0.3, 1); for 2024 (1, 0.9, 0, 0) with the most-deprived tier's share of the threat estimated, since many of them stood to gain from sub-classification; and for the EWS amendment the reform's own vector. Party backing is coded from reporting: 0.2 for 2018 (locally organized), 0.1 for September 2018, 0.6 for 2024 and 0 for 2019. It sets the party amplifier, with backing 0.6 corresponding to the reform scenario's  $a = 1.5$ . The magnitudes are relative to the reform's symbolic threat to SC agents, so a 2018 magnitude of 1 means that the reform threatens caste recognition as much as the 2018 ruling did.

## 7.3 Observation model

Expected reported events are  $\alpha \sum_s (P_s/10^5)^\gamma$ , where  $P_s$  is protester-days in state  $s$ , with Delhi's term multiplied by a media factor, since national media concentrate there (Hammond and Weidmann, 2014). The concave form means that a bandh spread over many states produces more reported events than a single rally of the same total size. The scale  $\alpha$ , exponent  $\gamma$  and Delhi factor are unknown. They are tied down by two anchor episodes with published crowd estimates: the Maratha march in Mumbai on 9 August 2017 (1 to 20 lakh person-days) and the Patidar agitation of August 2015 (3 to 20 lakh).

## 7.4 History matching

Fifteen unknowns get uniform priors:  $\bar{\theta}$  (5–8.5),  $\sigma_\theta$  (0.8–2.2),  $h$  (0.5–1), the four episode magnitudes, the 2024 deprived-tier share, the bandh-day death rate ( $10^{0.3}$  to  $10^3$  per crore protester-days), its dispersion, the three observation parameters and the two anchor crowd sizes. For each parameter set, every mechanism episode is simulated with two seeds. A set is *implausible* if, for any target, the standardized distance

$$I = \frac{|\log \hat{y} - \log y|}{\sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{Poisson}}^2 + \sigma_{\text{sim}}^2 + \sigma_{\text{disc}}^2}} \quad (4)$$

exceeds 3 (Andrianakis et al., 2015; Vernon et al., 2010), where the variances are observation error (from the precision interval), Poisson counting error, simulation noise between seeds, and a model discrepancy of 0.5 on the log scale. Deaths on the bandh day are compared on the natural scale with negative-binomial variance and a discrepancy of 2 deaths. Three waves of 6,000 sets are run; each later wave samples within the range of the previous wave's survivors. The sets that survive all waves form the not-ruled-out-yet (NROY) set used by the grounded specification.

## 7.5 Out-of-sample tests

Two kinds of test use data not used to fit.

1. **Leave one episode out.** For each of the three bandh episodes, a separate history match uses only the other episodes. Its survivors predict the held-out episode's bandh-day deaths and its distribution of protest across states; the held-out magnitude is not identified without that episode's data, so the prediction integrates over its prior. The predictions are written to files and hashed (SHA-256) before a separate script compares them with the observations.

2. **Spatial distribution.** Event counts by state are not used in calibration. For each episode, the Spearman correlation across large states between the model’s protester-days and the observed event counts is compared with two naive baselines: state population and state SC-plus-ST population.

## 8 Model development: what went wrong and how it was caught

The stylized model is the fourth specification. We report the earlier ones because the failures show which mechanisms matter and why validation must go beyond calibration. Every number in this section comes from `experiments/replay_model_development.py`, which re-runs each earlier specification with the current code.

**Iteration 1: unbounded contagion.** The first specification used linear social influence,  $S_{i,t} = 1.6 n_{c,t-1} + 25 v_{g,t-1}$ , narrow thresholds ( $\sigma_\theta = 0.55, \tau = 0.22$ ) and equal weight on material loss and symbolic threat ( $w_m = 1.0, w_s = 2.2$ ). At every mean threshold tried (2.8 to 3.3), it placed about 98.0 crore people on the street on the peak day, essentially every SC, ST and OBC agent. Even without contagion, the first bandh day would have drawn 30.6% of SC agents, against 0.70% in the final model; and linear contagion has no ceiling, so once a group’s national turnout reached a few percent the visibility term alone exceeded any threshold. The fix was to bound social influence with a saturating function.

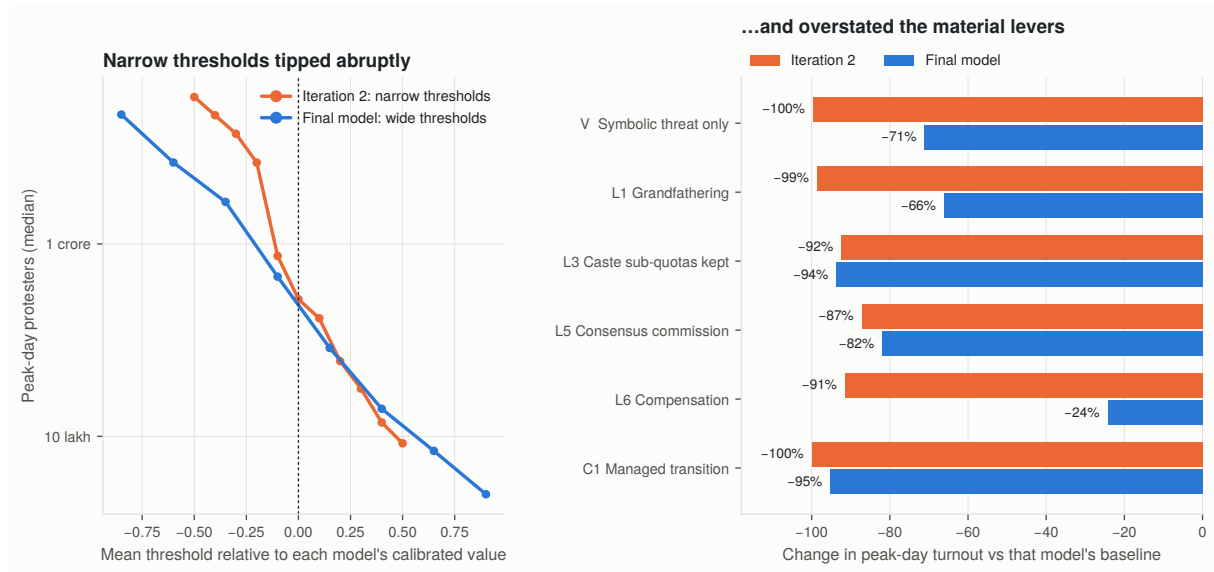
**Iteration 2: saturating contagion, narrow thresholds.** With saturating influence the model could be calibrated: at  $\bar{\theta} = 5.1$  it gave a baseline peak of 30.5 lakh. It had a knife-edge: turnout fell 3.1-fold between  $\bar{\theta} = 4.9$  (2.6 crore) and 5.0 (86.6 lakh), so any intervention that nudged grievance down appeared to eliminate protest. Grandfathering changed the peak by  $-99\%$  and compensation by  $-91\%$ . The decisive evidence came from a validation test: removing all material loss, the analogue of the 2018 ruling, left 10,000 protesters, 0.3% of the baseline. That contradicts S1. Iteration 2 had been calibrated to a plausible turnout, but for the wrong reason.

**Iteration 3: reweighting alone does not fix it.** Raising  $w_s$  and lowering  $w_m$ , recalibrating  $\bar{\theta}$  each time, left the symbolic-only shock below 5% of the baseline however the weights were set (Table 5). With a narrow threshold distribution, removing any component of grievance drops the population below the tipping point together.

**Table 5:** Iteration 3: reweighting symbolic and material grievance with narrow thresholds (6 runs each).

| $w_s$ | $w_m$ | $\bar{\theta}$ | Baseline peak (lakh) | Symbolic-only peak (lakh) | Symbolic-only share |
|-------|-------|----------------|----------------------|---------------------------|---------------------|
| 3.6   | 0.5   | 5.3            | 145.8                | 2.6                       | 1.8%                |
| 5.0   | 0.4   | 6.4            | 76.9                 | 2.7                       | 3.5%                |
| 6.0   | 0.3   | 7.1            | 95.8                 | 4.3                       | 4.5%                |

**Iteration 4: a wide threshold distribution.** Granovetter’s central point is that the distribution of thresholds, not only its mean, governs collective outcomes. Widening the spread to  $\sigma_\theta = 1.4$ , with  $\tau = 0.3$ , produced a smooth response: turnout falls about 2.3-fold per quarter-step of  $\bar{\theta}$  near the calibrated value. After recalibration ( $w_s = 3.3, w_m = 0.45, \bar{\theta} = 6.6$ ), the symbolic-only shock yields 29% of baseline turnout. Figure 1 compares iterations 2 and 4.



**Figure 1:** Left: peak turnout against mean threshold, centred on each model's calibrated value. Right: iteration 2 overstated material levers and understated symbolic-only protest.

**Lessons.** Calibration is not validation: iteration 2 matched the turnout target but could not reproduce a documented episode. Every positive feedback needs a bound. Threshold heterogeneity is a substantive assumption; it determines whether interventions look marginal or miraculous. And some parameters are not separately identified: only the product  $\lambda w_m$  enters grievance, and the ratio  $w_s/w_m$  is not identified by any turnout target (Section 11.6). The same lessons motivated the grounded specification: a single turnout target and a single validation episode are too weak a test, so the grounded model is confronted with several episodes, including a null episode and one in which the upper castes, not SCs and STs, mobilized.

## 9 Experimental design

Table 6 maps each scenario to its parameter changes. The reference values are those of the original model; Section 11.5 replaces each with a prior. In the grounded specification, material levers act on the segment table (grandfathering scales every loss by 0.3; compensation halves the losses of the three above-line losing segments), and L2 and L3 take their material changes from the allocation model (Section 5). In a package, allocation-rule levers are applied first, so that levers that scale losses act on the rule actually adopted.

**Table 6:** Scenarios and how they enter the model (reference values).

| Code | Scenario                                     | Parameter changes                                                                                                                           |
|------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Base | Abrupt ₹8L income-only switch                | None                                                                                                                                        |
| V    | Symbolic threat only (validation)            | All material changes set to 0                                                                                                               |
| L1   | Grandfather current cohorts, 10-year glide   | All material losses $\times 0.3$ ; $s_g \times 0.95$                                                                                        |
| L2   | Expand seats by a quarter                    | Stylized: below-line SC/ST/OBC losses set to 0. Grounded: allocation model with 25% more seats                                              |
| L3   | Keep caste quotas, income filter inside them | $s_g \times 0.4$ for SC, ST, OBC. Stylized: below-line losses 0. Grounded: allocation model with SC/ST quotas closed to above-line families |
| L4   | Sub-classify to favour the most-deprived     | Most-deprived SC/ST tier: material change reduced by 0.5 (half the unit loss); its symbolic threat falls from 0.8 to 0.3 times the group's  |
| L5   | Data-first commission, cross-party consensus | $a$ : 1.5 to 1.0; $s_g \times 0.8$                                                                                                          |
| L6   | Compensate above-line losers                 | Above-line losses $\times 0.5$                                                                                                              |
| L7   | Guarantee untouched protections              | $s_g \times 0.85$                                                                                                                           |
| B1   | Internet shutdowns                           | $\beta_n, \beta_v \times 0.6$ ; bandh mobilization $\times 0.8$ ; violent deaths $\times 2.5$                                               |
| B2   | Heavy policing and mass arrests              | Turnout cost 0.4 on bandh days; police-attributed deaths raised so total deaths $\times 3$ ; martyr response $\times 1.5$                   |
| C1   | Managed transition                           | L2+L1+L5+L6+L7                                                                                                                              |
| C2   | Hybrid package                               | L3+L4+L1+L5+L6+L7                                                                                                                           |

The main comparison runs every scenario 50 times with paired draws in the grounded specification, and 50 times in each of two regimes of the stylized specification (central, and a high-mobilization regime with  $\bar{\theta}$  lowered by 0.5). The outcomes are peak-day protesters, unique participants over the 40-day campaign, protester-days, and deaths. The high-mobilization regime is reported in Appendix E.

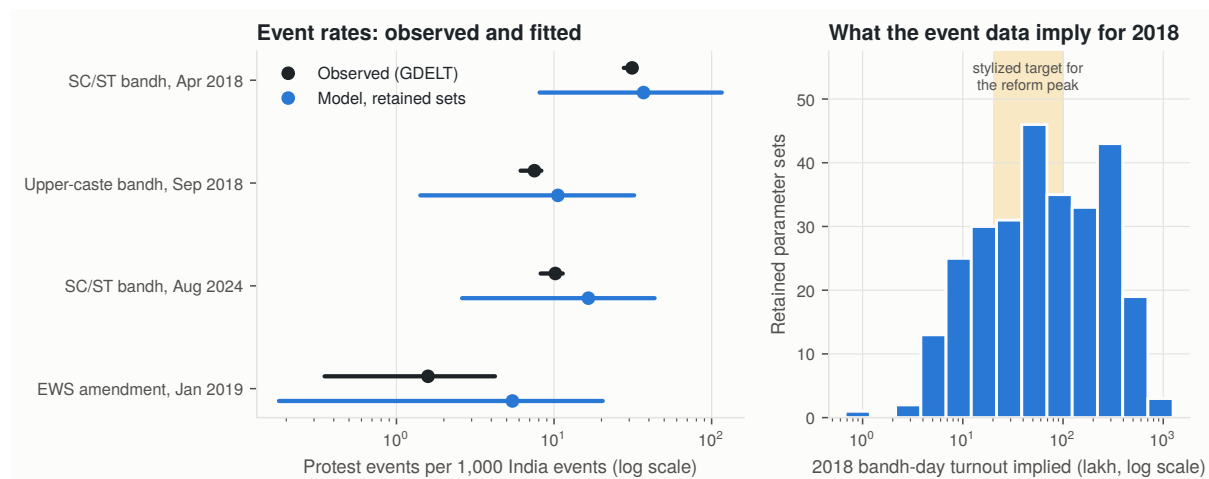
## 10 Calibration results

**What survives.** Of 6,000 parameter sets drawn from the priors, 19 were not ruled out by the event counts, deaths and anchors. After 3 waves, 281 sets survive (Figure 2). By construction, every survivor reproduces every target within the stated uncertainty, including the September 2018 upper-caste bandh, which the model generates from the same mechanisms with the threat pointing the other way. The fit to the EWS null episode is the loosest: the survivors' median event rate lies above the observed one, and the fit relies on the wide observation interval that the low precision of the EWS filter produces.

**What the data identify.** The mean threshold is 7.06 (90% interval 5.35–8.35), close to the stylized model's 6.6, which was tuned to an independent turnout target. The 2018 shock is 1.59 (0.50–2.38) times the stylized reform's symbolic threat. Relative to 2018, the 2024 shock has median 0.57 (0.06–1.89), the September 2018 upper-caste shock 0.52 (0.07–1.57), and the EWS amendment 0.43 (0.07–1.16). The medians order the episodes as reporting does, but every interval is wide. The share of single-group neighbourhoods is 0.78 (0.54–0.98): the data do not support complete segregation, but do not pin mixing down either.

**What they do not.** The implied turnout on 2 April 2018 is 63 lakh, with a 90% interval of 7–539 lakh. Event counts trade off against the observation model's scale and exponent, and the

two anchors narrow that trade-off only partly. The event data rule out the lowest reading of the press reports (“thousands”), but they cannot distinguish a few lakh from a few crore. This is the central limitation of calibrating to event data here: it constrains relative magnitudes far better than absolute turnout. The bandh-day death rate is 60 per crore protester-days (5–769), far above the stylized model’s ordinary-day rate of 4, because 11 to 14 people died on a single day in 2018.



**Figure 2:** Left: observed event rates (median and 90% interval from the precision correction) and the rates produced by the retained parameter sets (median and 90% range). Right: 2018 bandh-day turnout implied by the retained sets, with the stylized model’s calibration range for the reform peak shaded for reference.

**Out-of-sample tests.** The leave-one-episode-out predictions were committed to the public repository, with their hashes, before they were scored (Table 7). They are weak.

- **Deaths.** Without its own data, 2018’s bandh-day deaths are predicted at a median of 9 against 11–14 observed, but with an interval so wide as to be uninformative. For the two peaceful bandhs the model predicts deaths where none occurred. The model has one bandh-day death rate for all episodes, and the data say that violence is specific to the episode: it depends on policing, local counter-mobilization and chance in ways the model does not represent. The death outputs of the reform scenarios should be read in that light.
- **Geography.** Across 20 large states, the model’s distribution of protest correlates with the observed distribution of events at 0.57 for 2018 when that episode is held out, better than state population (0.38) and state SC-plus-ST population (0.49). For 2024 it roughly ties the SC-plus-ST baseline (0.73 against 0.72); for the upper-caste bandh it does worse than both baselines (0.42 against 0.60 and 0.63). Table 8 gives the same comparison with the full calibration. The district geography adds little predictive power beyond demography.

These results limit what the grounded specification can claim. It is grounded in the sense that its inputs come from data and its survivors reproduce four episodes’ event counts; it is not a validated forecasting model of where, how large or how deadly protest will be.

**Table 7:** Leave-one-episode-out predictions, frozen and hashed before scoring. Deaths: observed range and predicted median with 90% interval of the mean. Spatial: Spearman correlation across large states between predicted protester-days and observed events, against two naive baselines.

| Held-out episode                    | Sets | Deaths | Predicted         | Inside | Model | Population | SC+ST |
|-------------------------------------|------|--------|-------------------|--------|-------|------------|-------|
| SC/ST Bharat Bandh, 2 April 2018    | 320  | 11–14  | 9.0 (0.0–1429.3)  | yes    | 0.57  | 0.38       | 0.49  |
| SC/ST Bharat Bandh, 21 August 2024  | 361  | 0      | 24.3 (0.1–1453.8) | no     | 0.73  | 0.57       | 0.72  |
| Upper-caste bandh, 6 September 2018 | 373  | 0      | 18.8 (0.1–1678.8) | no     | 0.42  | 0.60       | 0.63  |

**Table 8:** Spatial test with the retained sets from the full calibration: Spearman correlation across large states between model protester-days and observed core-day events (median, 90% range over retained sets), against state population and state SC-plus-ST population.

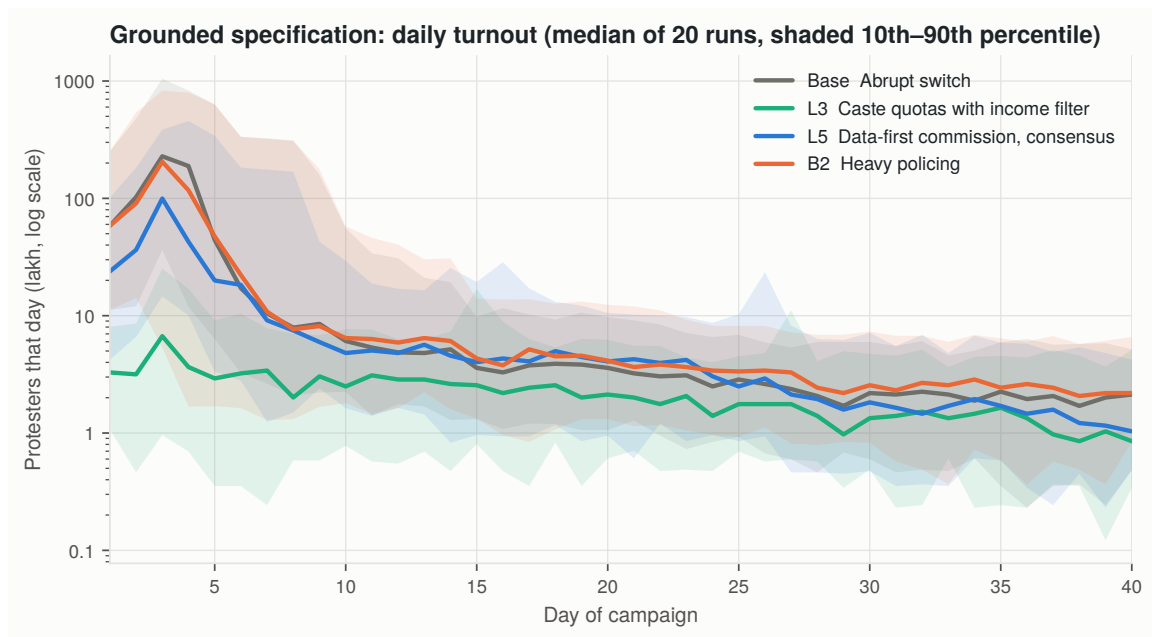
| Episode                             | Model               | Population | SC+ST population |
|-------------------------------------|---------------------|------------|------------------|
| SC/ST Bharat Bandh, 2 April 2018    | 0.52 (0.43 to 0.60) | 0.38       | 0.49             |
| SC/ST Bharat Bandh, 21 August 2024  | 0.65 (0.51 to 0.74) | 0.57       | 0.72             |
| Upper-caste bandh, 6 September 2018 | 0.44 (0.26 to 0.62) | 0.60       | 0.63             |

## 11 Results

### 11.1 The abrupt switch

In the grounded specification an abrupt switch produces a median peak of 2.14 crore (80% interval 30.6–1248.7 lakh), 3.32 crore participants and a median of 95 deaths. Organizations call the first bandh on day 3 in the median run, and the government concedes in 100% of runs, after a median of 5 days. The stylized specification, tuned to a peak of 20 lakh to 1 crore, gives 46.1 lakh.

The grounded level should not be read as a forecast. It is 1.59 times the stylized symbolic threat because that is what the 2018 episode implies, it inherits the wide range of 2018 turnouts that the event data allow, and it assumes that the reform shocks caste recognition as much as the 2018 ruling did ( $R = 1$ ) and that opposition parties back the protest. Section 12.1 shows that the level moves by more than an order of magnitude across these assumptions. The comparisons between interventions, which are paired within each world, are the object of the analysis.



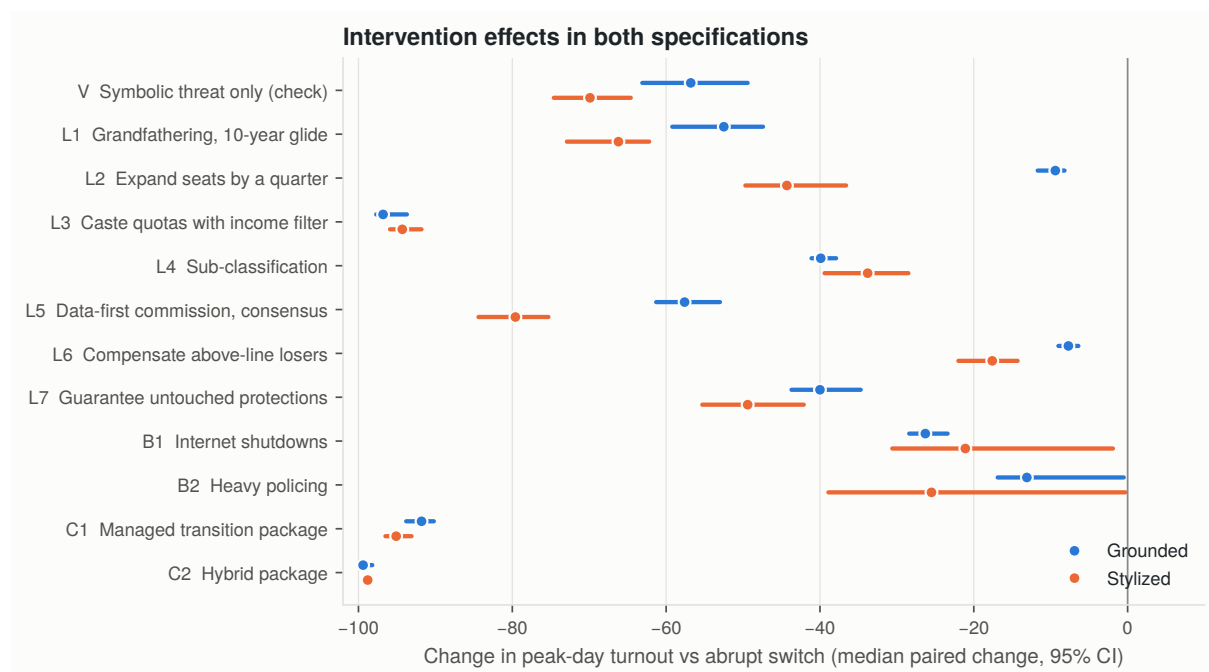
**Figure 3:** Grounded specification: daily turnout for four scenarios. Bandh days and concession are endogenous, so the median path is smoother than any single run.

## 11.2 Interventions

Table 9 reports every scenario in the grounded specification, and Figure 4 compares its paired effects with the stylized specification's.

**Table 9:** Grounded specification, 50 paired runs. Peak and interval (10th–90th percentile) in lakh; change is the median paired change in peak turnout with a 95% bootstrap interval; ever protested in crore; deaths are medians.

|      | Scenario                                           | Peak  | 10th–90th   | Change | 95% CI       | Ever | Deaths |
|------|----------------------------------------------------|-------|-------------|--------|--------------|------|--------|
| Base | Abrupt ₹8L income-only switch                      | 214.1 | 30.6–1248.7 | –      | –            | 3.32 | 95     |
| V    | Validation: symbolic threat only (2018-type shock) | 64.1  | 10.6–719.8  | –57%   | –63 to –49%  | 1.67 | 67.5   |
| L1   | Grandfather current cohorts + 10-year glide        | 83.5  | 13.6–751.2  | –52%   | –59 to –47%  | 1.69 | 70     |
| L2   | Expand seats by a quarter (EWS precedent)          | 193.4 | 27.1–1193.6 | –9%    | –12 to –8%   | 2.99 | 93.5   |
| L3   | Keep caste sub-quotas, add income filter           | 12.5  | 1.1–34.1    | –97%   | –98 to –94%  | 0.39 | 10     |
| L4   | Sub-classify to favour most-deprived               | 127.7 | 25.6–751.4  | –40%   | –41 to –38%  | 2.23 | 99.5   |
| L5   | Data-first commission + cross-party consensus      | 84.4  | 11.8–700.9  | –58%   | –61 to –53%  | 1.57 | 57.5   |
| L6   | Compensate above-line losers                       | 197.0 | 28.2–1204.4 | –8%    | –9 to –6%    | 3.00 | 95     |
| L7   | Guarantee untouched protections                    | 114.0 | 22.2–846.8  | –40%   | –44 to –35%  | 1.92 | 78.5   |
| B1   | Internet shutdowns                                 | 146.5 | 26.5–793.8  | –26%   | –28 to –23%  | 2.60 | 139    |
| B2   | Heavy policing and mass arrests                    | 188.5 | 24.4–1167.5 | –13%   | –17 to 0%    | 3.41 | 219.5  |
| C1   | Managed transition package (L1+L2+L5+L6+L7)        | 18.7  | 1.9–78.6    | –92%   | –94 to –90%  | 0.70 | 15     |
| C2   | Hybrid design package (L3+L4+L1+L5+L6+L7)          | 1.7   | 0.4–12.5    | –99%   | –100 to –98% | 0.13 | 2      |



**Figure 4:** Median paired change in peak-day turnout with 95% bootstrap intervals, grounded and stylized (central regime) specifications.

**Keeping caste as the qualifier dominates.** Keeping caste quotas and adding an income filter inside them (L3) changes the grounded peak by  $-97\%$  (95% CI  $-98$  to  $-94\%$ ), and the stylized peak by  $-94\%$ . In the grounded specification the lever does two things: it lowers symbolic threat, and, according to the allocation model, it turns the reform's large seat loss for below-line SC and ST candidates into a gain (Section 5). This is also the lever that least resembles the reform: it keeps caste as a criterion, so it changes the question more than it answers it.

**Among levers that keep the income-only end state, process and phasing come first.** A data-first commission with cross-party consensus (L5) changes the peak by  $-58\%$  ( $-61$  to  $-53\%$ ), and grandfathering with a ten-year glide (L1) by  $-52\%$  ( $-59$  to  $-47\%$ ). Their order is close in the grounded specification, and Section 11.5 shows that it depends on the priors over how strongly each lever acts. Statutory guarantees (L7,  $-40\%$ ) and sub-classification (L4,  $-40\%$ ) follow.

**Seat expansion and compensation do little.** Expanding seats by a quarter (L2) changes the grounded peak by only  $-9\%$ , against  $-44\%$  in the stylized specification. The difference is the allocation model: the stylized specification assumed that expansion removes the below-line loss, while the allocation model shows that a merged pool with a quarter more seats still takes most of their seats from below-line SC and ST candidates ( $-63\%$  for SC). The EWS precedent does not transfer, because EWS added a category without merging any. Compensating above-line losers (L6) is weakest ( $-8\%$ ), because it reaches a small minority and does not touch symbolic threat.

**Packages.** The managed transition (C1) keeps the income-only end state and changes the peak by  $-92\%$  ( $-94$  to  $-90\%$ ); the hybrid package (C2) by  $-99\%$ .

**Material grievance matters more than the stylized model assumed.** Removing all material change (V) changes the grounded peak by  $-57\%$ , against  $-70\%$  in the stylized specification. The allocation model makes the material loss larger and broader than the stylized specification assumed, because below-line SC and ST candidates lose most of their seats, not a small share. Symbolic threat is still the larger driver, but not by as much.

### 11.3 Suppression

Internet shutdowns (B1) change the grounded peak by  $-26\%$  and multiply deaths by 1.4; heavy policing (B2) changes it by  $-13\%$  ( $-17$  to  $0\%$ ) and multiplies deaths by 2.6. Part of each death ratio is an input: B1 assumes 2.5 times as many violent deaths and B2 three times as many police-attributed deaths. Section 11.4 separates the two. Unlike the stylized specification, where heavy policing raises cumulative participation by  $+14\%$ , the grounded specification shows a change of  $-4\%$ : campaigns end in concession within days, so the martyr effect has little time to act. Whether suppression backfires on participation is therefore not a robust result; that it costs lives is, since it follows from the assumed rise in deaths.

### 11.4 What is input and what is emergent

Every lever changes more than one parameter. Re-running levers with one channel at a time (stylized specification, same paired worlds) separates them.

- **L5.** Lowering the party amplifier alone changes the peak by  $-39\%$ ; lowering symbolic threat alone by  $-60\%$ . The commission's effect comes mostly from its assumed 20% cut in symbolic threat, not from withdrawing party machinery. Given S3 (2018 was large

without party backing), that is the more defensible channel, but it is also the less certain one.

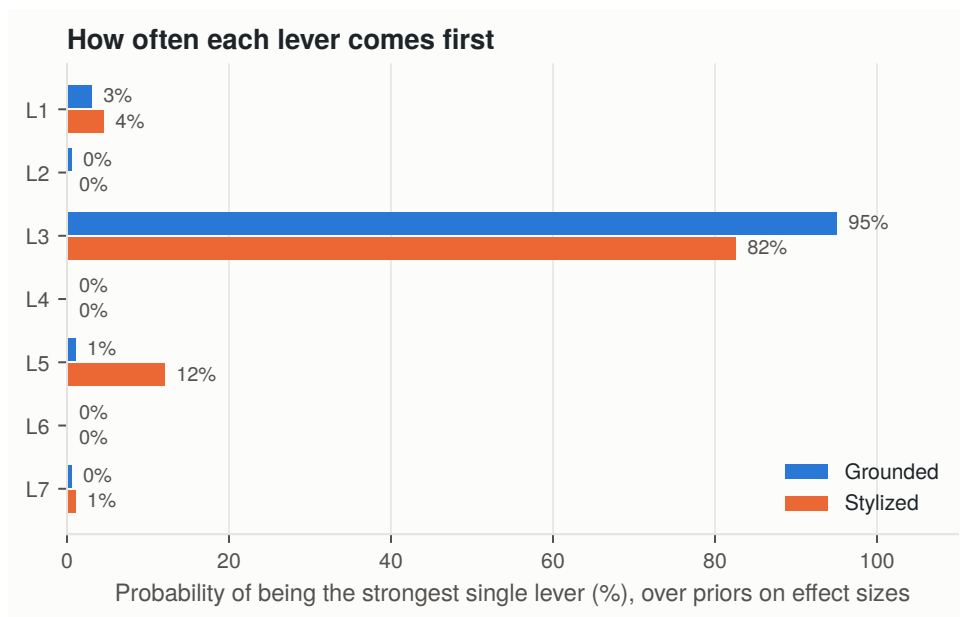
- **L3.** Its symbolic channel alone gives  $-91\%$  and its material channel alone  $-44\%$ .
- **L1.** Its material channel gives  $-57\%$  and its 5% symbolic cut  $-17\%$ .
- **B2.** The turnout cost alone changes the peak by  $-51\%$ ; the extra violence and martyr response alone raise it by  $+42\%$ . The full scenario multiplies deaths by 3.9, of which the assumed multiplier of 3 is input; the emergent part, a factor of 1.3, comes from the change in protester-days ( $+20\%$ ).

### 11.5 Uncertainty over how interventions act

The reference effect sizes in Table 6 are judgements. Each was given a uniform prior (for example, the commission's cut in symbolic threat from 0 to 30% and its cut in the party amplifier from 10 to 50%; the full list is in `protest_simulation/intervention_priors.py`). Each run draws one set of effect sizes and applies it to every lever in that run's world. Table 10 gives, for each specification, the paired effect and the probability that each lever ranks first to last; Figure 5 summarizes the probability of ranking first.

**Table 10:** Lever effects and rank probabilities under priors over effect sizes (200 runs, grounded specification; stylized in Appendix D). Change: median paired change in peak with 95% CI; rank 1 = largest reduction.

| Lever | Change  | 95% CI           | 1   | 2   | 3   | 4   | 5   | 6   | 7   |
|-------|---------|------------------|-----|-----|-----|-----|-----|-----|-----|
| L1    | $-58\%$ | $-61$ to $-55\%$ | 3%  | 48% | 30% | 12% | 6%  | 0%  | 1%  |
| L2    | $-7\%$  | $-8$ to $-6\%$   | 0%  | 0%  | 1%  | 2%  | 8%  | 44% | 44% |
| L3    | $-89\%$ | $-92$ to $-87\%$ | 95% | 2%  | 1%  | 0%  | 0%  | 0%  | 1%  |
| L4    | $-33\%$ | $-35$ to $-32\%$ | 0%  | 7%  | 15% | 38% | 36% | 2%  | 1%  |
| L5    | $-42\%$ | $-48$ to $-39\%$ | 1%  | 28% | 34% | 16% | 16% | 2%  | 2%  |
| L6    | $-8\%$  | $-8$ to $-7\%$   | 0%  | 2%  | 1%  | 3%  | 10% | 46% | 38% |
| L7    | $-32\%$ | $-35$ to $-27\%$ | 0%  | 12% | 18% | 28% | 24% | 4%  | 12% |



**Figure 5:** Probability that each single lever gives the largest reduction in peak turnout, over the priors on effect sizes.

L3 is the strongest single lever with probability 95% in the grounded specification and 82% in the stylized one. The commission beats grandfathering with probability 34% in the grounded specification and 58% in the stylized one. The reference values in Table 6 sit near the strong end of the commission's prior, which is why it ranks above grandfathering in Table 9; averaged over the priors, grandfathering is the stronger lever more often in the grounded specification. The order of the two is not settled by the model. Compensation is in the bottom two in 84% of grounded draws and 94% of stylized draws; in the grounded specification seat expansion shares the bottom (88%).

For suppression, the prior on the response of turnout to deaths runs from deterrence to strong backfire. Heavy policing raises cumulative participation in 22% of grounded runs and 37% of stylized runs; in the stylized specification the share rises from 0% in the quarter of runs with the weakest response to 74% in the quarter with the strongest. Backfire on participation is therefore a consequence of the assumed response to deaths, which no episode identifies.

## 11.6 Material versus symbolic weight

The ratio of the symbolic to the material grievance weight is not identified by any turnout target, and the episodes involve no material change, so they do not identify  $w_m$  either. In the stylized specification, a sweep that holds  $w_s w_m$  fixed and recalibrates  $\bar{\theta}$  at each point finds that a material lever is strongest only when  $w_s/w_m$  is at most 1.47, against the reference 7.3; the 2018-type validation check admits ratios 3.67–22.00. In the grounded specification, where  $\bar{\theta}$  comes from the episodes, we multiply  $w_m$  directly (Table 11). No material lever becomes the strongest at any multiplier up to 8. Grandfathering overtakes the commission once  $w_m$  is doubled, and compensation stops being the weakest lever once  $w_m$  is quadrupled; grandfathering approaches L3 only at the largest multiplier, where the baseline peak is implausibly large. A heavier material weight therefore reorders the middle of the ranking, not its top.

**Table 11:** Grounded specification: paired change in peak by lever as the material weight is multiplied ( $R = 1$ , backing 0.6).

| $w_m$ multiplier | Baseline peak | L1   | L2   | L3   | L5   | L6   | L7   | First |
|------------------|---------------|------|------|------|------|------|------|-------|
| 0.25             | 88.3 lakh     | –26% | –2%  | –94% | –63% | –2%  | –44% | L3    |
| 0.5              | 1.23 crore    | –35% | –5%  | –95% | –62% | –4%  | –42% | L3    |
| 1                | 2.14 crore    | –52% | –9%  | –97% | –58% | –8%  | –40% | L3    |
| 2                | 3.94 crore    | –71% | –13% | –96% | –43% | –13% | –29% | L3    |
| 4                | 15.49 crore   | –79% | –13% | –92% | –25% | –14% | –14% | L3    |
| 8                | 36.14 crore   | –86% | –10% | –89% | –6%  | –14% | –3%  | L3    |

## 12 Robustness

### 12.1 The grounded specification's unidentified assumptions

Table 12 varies the reform's shock relative to 2018 ( $R = 0.5, 1, 1.5$ ) and the party backing of the protest (0.2, the 2018 level, and 0.6, the 2024 level). The baseline peak ranges from 29.0 lakh to 6.78 crore. L3 is the strongest single lever in 6 of 6 cells, and compensation is the weakest in 6.

**Table 12:** Grounded specification under alternative values of the reform's shock ( $R$ ) and party backing; 50 paired runs per cell.

| $R$ | Backing | Peak       | Deaths | Conceded | L1   | L3   | L5   | L6   | First |
|-----|---------|------------|--------|----------|------|------|------|------|-------|
| 0.5 | 0.2     | 29.0 lakh  | 25     | 88%      | −66% | −88% | −53% | −11% | L3    |
| 0.5 | 0.6     | 37.0 lakh  | 28     | 88%      | −64% | −90% | −57% | −11% | L3    |
| 1.0 | 0.2     | 2.06 crore | 92.5   | 98%      | −53% | −97% | −56% | −8%  | L3    |
| 1.0 | 0.6     | 2.14 crore | 95     | 100%     | −52% | −97% | −58% | −8%  | L3    |
| 1.5 | 0.2     | 6.09 crore | 182.5  | 100%     | −43% | −97% | −48% | −5%  | L3    |
| 1.5 | 0.6     | 6.78 crore | 209    | 100%     | −42% | −97% | −49% | −5%  | L3    |

## 12.2 Structural ensemble

Each variant in Table 13 switches on one structural alternative, or all of them, recalibrates  $\bar{\theta}$  so that the stylized baseline again hits its turnout target, and reruns every lever with 100 paired runs. L3 is strongest in 11 of 11 variants and compensation weakest in 11; L5 beats L1 in 11. Heavy policing raises cumulative participation in 10 variants, and not when deaths deter.

**Table 13:** Structural ensemble (stylized specification, recalibrated): paired change in peak for four levers, change in cumulative participation under heavy policing, and the strongest / weakest single lever.

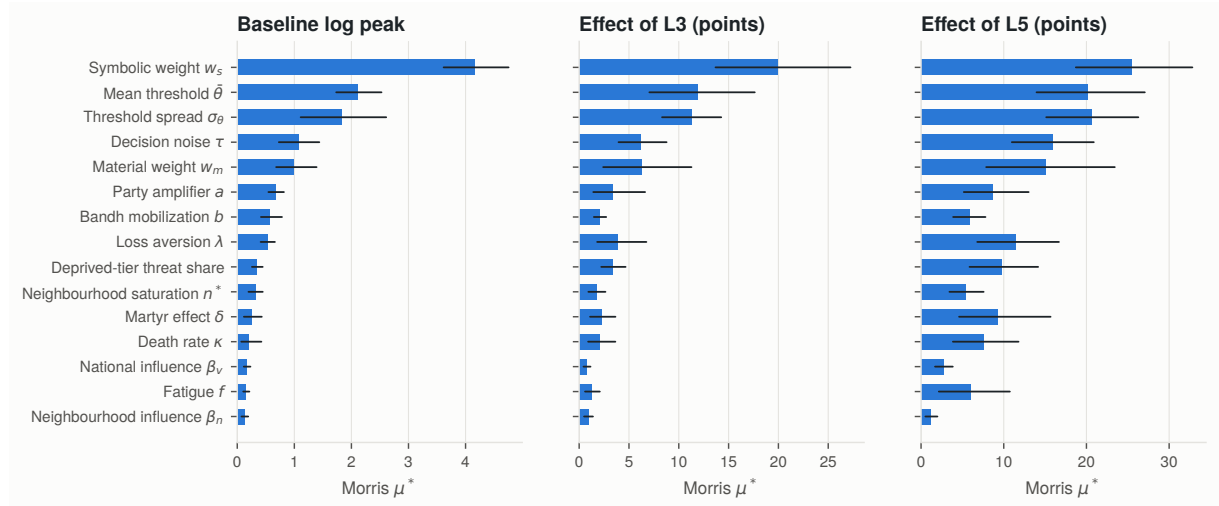
| Variant                                                                                 | $\bar{\theta}$ | L1   | L3   | L5   | L6   | B2 cum. | First / last |
|-----------------------------------------------------------------------------------------|----------------|------|------|------|------|---------|--------------|
| Reference structure                                                                     | 6.60           | −69% | −95% | −81% | −18% | +14%    | L3 / L6      |
| Separate random streams for decisions, deaths and events                                | 6.64           | −72% | −94% | −82% | −21% | +17%    | L3 / L6      |
| Logistic (heavier-tailed) threshold distribution                                        | 6.93           | −49% | −82% | −63% | −12% | +7%     | L3 / L6      |
| 3% activist core with low thresholds                                                    | 7.05           | −60% | −94% | −74% | −12% | +15%    | L3 / L6      |
| Organizations call bandhs when turnout is credible                                      | 6.67           | −81% | −98% | −86% | −20% | +10%    | L3 / L6      |
| Government may concede; General category counter-mobilizes                              | 6.54           | −62% | −93% | −73% | −18% | +5%     | L3 / L6      |
| Negative-binomial deaths, split into police-attributed and other                        | 6.63           | −68% | −94% | −78% | −19% | +18%    | L3 / L6      |
| Deaths deter rather than mobilize                                                       | 6.24           | −54% | −88% | −68% | −15% | −40%    | L3 / L6      |
| 30% of neighbourhoods mixed across groups                                               | 6.59           | −68% | −95% | −80% | −18% | +14%    | L3 / L6      |
| Half of above-line OBC non-creamy today; 15% of below-line General fail EWS asset tests | 6.61           | −69% | −94% | −81% | −19% | +18%    | L3 / L6      |
| All alternatives above except deterrence                                                | 6.95           | −54% | −97% | −65% | −15% | +9%     | L3 / L6      |

## 12.3 Global sensitivity

Morris screening over fifteen behavioural parameters (20 trajectories, 320 design points, stylized specification) finds that the symbolic weight dominates the baseline peak and every lever's effect, followed by the mean threshold and the threshold spread (Table 14, Figure 6). Across all design points, L3 cuts the peak more than L5 in 98% and L5 more than L1 in 75%.

**Table 14:** Morris screening: mean absolute elementary effect  $\mu^*$  on the baseline log peak and on the paired peak change (percentage points) of L1, L3 and L5, and  $\sigma$  for the log peak.

| Parameter                         | Log peak | L1 | L3 | L5 | $\sigma$ (log peak) |
|-----------------------------------|----------|----|----|----|---------------------|
| Symbolic weight $w_s$             | 4.16     | 26 | 20 | 25 | 1.38                |
| Mean threshold $\bar{\theta}$     | 2.11     | 24 | 12 | 20 | 0.94                |
| Threshold spread $\sigma_\theta$  | 1.83     | 26 | 11 | 21 | 1.73                |
| Decision noise $\tau$             | 1.08     | 17 | 6  | 16 | 0.81                |
| Material weight $w_m$             | 0.99     | 28 | 6  | 15 | 0.82                |
| Party amplifier $a$               | 0.68     | 9  | 3  | 9  | 0.32                |
| Bandh mobilization $b$            | 0.57     | 8  | 2  | 6  | 0.44                |
| Loss aversion $\lambda$           | 0.53     | 15 | 4  | 11 | 0.29                |
| Deprived-tier threat share        | 0.34     | 10 | 3  | 10 | 0.23                |
| Neighbourhood saturation $n^*$    | 0.32     | 5  | 2  | 5  | 0.31                |
| Martyr effect $\delta$            | 0.26     | 9  | 2  | 9  | 0.43                |
| Death rate $\kappa$               | 0.20     | 8  | 2  | 8  | 0.46                |
| National influence $\beta_v$      | 0.17     | 4  | 1  | 3  | 0.14                |
| Fatigue $f$                       | 0.16     | 3  | 1  | 6  | 0.13                |
| Neighbourhood influence $\beta_n$ | 0.13     | 2  | 1  | 1  | 0.13                |

**Figure 6:** Morris  $\mu^*$  with 95% bootstrap intervals over trajectories.

## 12.4 One-at-a-time sensitivity, L3's symbolic assumption, and convergence

Each of twelve parameters perturbed by  $\pm 25\%$  (or  $\pm 0.25$  for  $\bar{\theta}$ ) leaves the ranking of levers largely intact: the Spearman correlation with the reference ranking has a median of 1.00 and a minimum of 0.93, and L3 is strongest in 24 of 24 perturbations (Table 15). L3 rests on the assumption that retaining caste categories removes 60% of symbolic threat; the 2024 bandh, triggered by a suggested SC/ST creamy layer, is reason for caution. Leaving 40%, 60% and 80% of the threat in place gives stylized peaks of 2.7 lakh, 4.6 lakh and 9.2 lakh. Medians show no systematic trend with agent count between 30,000 and 240,000 agents (Table 16).

**Table 15:** One-at-a-time sensitivity (stylized). Peaks in lakh (20 runs);  $\rho$  for lowered and raised parameter; strongest and weakest single lever (lowered/raised).

| Parameter                     | Lowered | Raised | $\rho_-$ | $\rho_+$ | Strongest | Weakest |
|-------------------------------|---------|--------|----------|----------|-----------|---------|
| Loss aversion                 | 32.4    | 85.1   | 0.98     | 0.98     | L3/L3     | L6/L4   |
| Material-loss weight          | 32.4    | 85.1   | 0.98     | 0.98     | L3/L3     | L6/L4   |
| Symbolic-threat weight        | 10.6    | 230.1  | 0.98     | 0.98     | L3/L3     | L6/L6   |
| Opposition-party amplifier    | 26.8    | 78.1   | 1.00     | 1.00     | L3/L3     | L6/L6   |
| Neighbourhood influence       | 43.5    | 46.2   | 1.00     | 1.00     | L3/L3     | L6/L6   |
| National visibility influence | 42.5    | 46.5   | 1.00     | 1.00     | L3/L3     | L6/L6   |
| Threshold spread              | 5.8     | 222.3  | 0.98     | 0.93     | L3/L3     | L4/L6   |
| Decision noise                | 30.0    | 84.9   | 1.00     | 1.00     | L3/L3     | L6/L6   |
| Fatigue                       | 56.4    | 40.6   | 1.00     | 1.00     | L3/L3     | L6/L6   |
| Death rate                    | 37.5    | 69.8   | 1.00     | 1.00     | L3/L3     | L6/L6   |
| Martyr effect per death       | 48.0    | 46.7   | 1.00     | 1.00     | L3/L3     | L6/L6   |
| Mean threshold ( $\pm 0.25$ ) | 117.0   | 18.9   | 1.00     | 1.00     | L3/L3     | L6/L6   |

**Table 16:** Agent-count convergence (stylized; 20 paired runs each).

| Agents  | People per agent | Baseline peak (lakh) | 10th–90th  | L3 peak (lakh) |
|---------|------------------|----------------------|------------|----------------|
| 30,000  | 48,700           | 53.0                 | 11.1–119.6 | 2.4            |
| 60,000  | 24,300           | 50.1                 | 10.7–122.2 | 1.9            |
| 120,000 | 12,200           | 42.7                 | 14.2–109.4 | 2.6            |
| 240,000 | 6,100            | 53.7                 | 11.2–120.0 | 2.3            |

## 13 Discussion

**What the revision changed.** Three findings of the original analysis did not survive grounding. First, the reform is not a small eligibility change: under the actual creamy-layer and EWS rules, about 6.8% of Indians lose eligibility, and in IIT admissions below-line SC and ST candidates, who keep eligibility, lose most of their seats to below-line General candidates. Second, seat expansion on the EWS model does little, because a merged pool still ranks SC and ST candidates against General candidates; the original analysis assumed the opposite. Third, that heavy policing raises total participation depends on an assumed response of turnout to deaths, and does not hold in the grounded specification, where campaigns end in concession quickly. Each of these changes came from replacing an assumption with data or with an explicit prior.

**What survived.** The ordering at the top and bottom of the lever ranking survived every analysis in this paper: keeping caste as the qualifier with an income filter inside it (L3) cuts protest most, and compensating the direct losers (L6) is among the least effective, joined in the grounded specification by seat expansion (L2). L3 is first in both specifications, across the priors over effect sizes, across the structural variants, across the Morris design, and across the grounded specification's unidentified shock size, party backing and material weight. Compensation is last or second to last in almost all of them; the exception is a material weight at least four times its reference value, which implies turnouts far beyond any recorded protest. The reason is structural. Most of the mobilizable grievance is symbolic and spread across the SC and ST population, and in the grounded specification the material loss is also concentrated on below-line SC and ST candidates; L3 removes much of both, while compensation reaches a small minority and touches neither.

**What did not settle.** Among levers that keep the income-only end state, the order of the data-first commission (L5) and grandfathering (L1) is not settled. At the reference values the commission is ahead; averaged over the priors on how strongly each lever acts, grandfathering is ahead more often (the commission wins with probability 34%). The order depends on how strongly a commission would lower symbolic threat, on the size of the reform's shock (grandfathering comes first when  $R = 0.5$ ), and on the material weight (grandfathering comes first once  $w_m$  is doubled), and none of these is identified. The decomposition shows that most of L5's effect in the model comes from its assumed cut in symbolic threat. Given that the 2018 bandh was large without party backing, the claim that consensus works by withholding party machinery is the weaker half of the mechanism. A commission's value, if it has one, would lie in changing what the reform means to those it affects.

**Compensation and the retrenchment literature.** Pierson (1994) treats compensation as a blame-avoidance tool. Its value depends on how grievance is composed. Where losses are material and the losers are the mobilizers, compensation addresses the grievance. Where the grievance is largely symbolic, or where the material loss falls on a broad group that compensation schemes do not reach, it does not.

**Policy implications.** Conditional on the model:

1. Whether caste stays a criterion matters more than any compensation scheme. An income filter inside caste categories achieves most of the income targeting that motivates the reform, at a small fraction of the unrest, and in IIT admissions it moves seats towards below-line SC and ST candidates rather than away from them.
2. A merged income-only pool is not a neutral re-labelling. It moves most reserved seats of below-line SC and ST candidates to below-line General candidates, and expanding capacity on the EWS model does not undo that.
3. If the income-only end state is the goal, phasing (grandfathering) and a process that changes what the reform signals (a commission, statutory guarantees) matter most, and they compound.
4. Suppression trades a lower peak for more deaths. Whether it also broadens protest depends on how deaths are received, which the model cannot tell.

**Feasibility.** The levers that work best are also the hardest to obtain or the furthest from the reform as proposed. Cross-party consensus on reservation is rare; L3 keeps caste as a criterion. The model treats political feasibility as exogenous.

## 14 Limitations and threats to validity

- **The reform's shock size is not identified.** The episodes identify how strongly the 2018 ruling threatened SC and ST recognition relative to the other episodes, not how strongly an income-only reform would. The grounded analysis assumes  $R = 1$  and backs the reform with opposition parties at the 2024 level. Table 12 shows that the level of protest moves by an order of magnitude across these assumptions; the ranking moves much less.
- **The material weight is not identified.** The episodes involve no material change, so  $w_m$  is assumed in both specifications. The break-even analyses bound how far it would have to move for the ranking to change.
- **Event counts are an indirect measure.** GDELT counts reports, not people. The observation model is concave and anchored on two episodes with published crowd estimates,

but the implied turnout for 2018 remains wide (7–539 lakh). The relevance labels were assigned from URL text by a single coder and have not been independently double-coded; the precision correction inherits that weakness. ACLED would provide an independent cross-check.

- **Party backing and shock direction are coded, not estimated.** Each episode’s party backing and the direction of its symbolic threat are judgements from reporting, held fixed in calibration; errors are absorbed by the estimated magnitudes.
- **Intervention mappings are priors.** How much a commission, a guarantee or a filter reduces threat is not observed. The rank probabilities integrate over wide priors, but they cannot correct a mapping whose whole range is wrong.
- **The allocation model covers one institution type.** Seat changes are estimated for IIT admissions, where rank lists are public. Government jobs, state institutions and medical admissions may differ; NEET failed our tail check. Family income is not observed per candidate and enters through priors constrained by aggregate evidence. Horizontal reservations and programme preferences are not modelled.
- **Omitted structure.** Jati-level heterogeneity beyond two SC/ST tiers, state-level variation in income and organization, media dynamics, courts and electoral feedback are not modelled. State government alignment was coded (`data/derived/state_government_alignment.csv`) but is not used by the model.
- **Generic dynamics.** Threshold cascades can produce heavy-tailed outcomes whatever their micro-foundations (Thron and Jackson, 2015). The episode tests, the structural ensemble and the separation of inputs from emergent effects are partial answers; they do not establish that the mechanisms are the right ones.

## 15 Ethics statement

The study uses no individual-level data. The population is synthetic and built from published aggregate shares; the JEE rank lists are used without roll numbers or names, and only rank and category are retained. Suppression tactics are included only to test whether they backfire, and they are not presented as options. Obfuscation, which is manipulating what citizens know about a reform, is deliberately excluded from the intervention set. The model is not a forecast of any actual policy. It is intended to inform the design of reforms that respect democratic contestation, not to help any actor suppress it. The state-level results describe where protest would concentrate; they are not, and should not be used as, a basis for pre-emptive policing.

## 16 Reproducibility

All code, derived data, results, figures and videos are available at [github.com/utsavbalhara/reservation-reform-protest-model](https://github.com/utsavbalhara/reservation-reform-protest-model). Seeds are fixed, and runs distributed across processes give results identical to serial runs. Every number in this paper is generated from the result files into LaTeX macros; none is typed by hand. A test suite checks the reference model against stored outputs and against an independent scalar reimplementations, and checks model properties (turnout moves in the expected direction with grievance and threshold, campaign accounting identities hold, each intervention touches only its own parameters, parallel and serial runs agree), and checks the choice rules, the rank-list extraction and the district population against their sources. The main commands, in order, are:

```
python -m data_pipelines.jee_advanced_rank_lists      # rank lists -> data/derived
python -m data_pipelines.gdelt_download                # GDELT daily files (raw, not stored)
python -m data_pipelines.gdelt_episode_events
python -m data_pipelines.episode_targets
```

```
python -m experiments.eligibility_accounting
python -m experiments.merged_pool_allocation --draws 300
python -m experiments.episode_calibration --wave-size 6000 --waves 3
python -m experiments.score_frozen_predictions
python -m experiments.compare_interventions --specification grounded
python -m experiments.compare_interventions --regime central
python -m experiments.compare_interventions --regime high-mobilization
python -m experiments.intervention_mapping_uncertainty --specification grounded
python -m experiments.intervention_mapping_uncertainty --specification stylized
python -m experiments.material_symbolic_break_even --specification stylized
python -m experiments.decompose_levers
python -m experiments.global_sensitivity
python -m experiments.structural_ensemble
python -m experiments.sensitivity_analysis
python -m experiments.check_robustness
python -m experiments.replay_model_development
python -m visualization.write_latex_result_macros
python -m visualization.write_paper_macros
python -m visualization.write_grounded_macros
python -m pytest
```

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## Cases and statutes

*Indra Sawhney v. Union of India*, 1992 Supp (3) SCC 217. *Ashoka Kumar Thakur v. Union of India*, (2008) 6 SCC 1. *Subhash Kashinath Mahajan v. State of Maharashtra*, (2018) 6 SCC 454. *Janhit Abhiyan v. Union of India*, Supreme Court of India, 7 November 2022. *State of Punjab v. Davinder Singh*, Supreme Court of India, 1 August 2024. The Constitution (One Hundred and Third Amendment) Act, 2019. The Scheduled Castes and the Scheduled Tribes (Prevention of Atrocities) Amendment Act, 2018. Brazil, Law No. 12.711 of 29 August 2012.

## A Novelty search

Before claiming novelty we searched Google Scholar (18 queries combining “reservation”, “caste”, “quota”, “affirmative action”, “Bharat Bandh”, “Mandal” and “Dalit” with “agent-based”, “simulation”, “threshold model”, “computational model” and “protest”; first page of results each) and SSRN (through a web search restricted to ssrn.com, since SSRN’s own search refused automated access). The nearest hits were agent-based models of food riots, of ethnic minority rule and civil war, and of affirmative action in hiring and college sorting, and a model of reservation policy in education markets; none modelled protest against reservation or compared reform designs by the mobilization they would provoke. OpenAlex and Semantic Scholar were rate-limited and not used. These were keyword searches, not a systematic review. The queries, hit counts and titles are in docs/literature\_search.md and docs/literature\_search\_scholar\_results.json.

## B Data sources and processing

- **JEE (Advanced) rank lists.** The Joint Implementation Committee reports (`jeeadv.ac.in/reports/<year>.pdf`) print the Common Rank List and each category’s rank list, sorted by roll number. Joining them on roll number gives the category of every candidate on the Common Rank List. Only rank and category are kept. The 2019 and 2020 lists fail a check that category ranks increase with Common Rank List rank and are not used. Registered, appeared and allotted counts by category are transcribed from the same reports; the 2024 appeared counts are imputed.
- **Census 2011.** District population and SC and ST counts from the Primary Census Abstract (640 districts; the ten Telangana districts are separated from Andhra Pradesh).
- **NFHS-5 (2019–21).** Caste of household head by state (India Report, Table 2.10), used for the OBC share of the non-SC, non-ST population.
- **GDELT 1.0.** Daily event files for each episode’s window; CAMEO root code 14, action located in India. Keyword filters and the exclusion keywords for each episode are in `data_pipelines/episodes.py`. The raw files are not stored in the repository; the pipeline downloads them.

- **Relevance audit.** 246 matched source URLs, sampled from core and non-core days of every episode, were labelled as relevant or not from the article’s URL text (headline slug), without opening the articles. Labels from URL text are a weaker basis than reading each article, and a second independent coding with an agreement statistic has not yet been done. The labels are in `data/derived/gdelt_relevance_audit.csv`.
- **Deaths and party backing.** Coded per episode from the cited reporting in `data_pipelines/episodes.py`.

## C Allocation model: robustness

**Table 17:** Median seat change by segment across processing orders and pool sizes, and for the two allocation-rule levers (reserved-first order).

| Order                           | Pool  | SC<br>above | SC<br>below | ST<br>below | OBC<br>below | GEN-EWS | GEN<br>below |
|---------------------------------|-------|-------------|-------------|-------------|--------------|---------|--------------|
| reserved-first                  | 59.5% | −96%        | −72%        | −85%        | +23%         | +71%    | +158%        |
| reserved-first                  | 49.5% | −95%        | −76%        | −87%        | +8%          | +43%    | +157%        |
| reserved-first                  | 69.5% | −98%        | −68%        | −82%        | +39%         | +101%   | +156%        |
| open-first                      | 59.5% | −94%        | −68%        | −82%        | +12%         | +51%    | +224%        |
| open-first                      | 49.5% | −92%        | −71%        | −83%        | +4%          | +41%    | +208%        |
| open-first                      | 69.5% | −96%        | −65%        | −80%        | +20%         | +62%    | +239%        |
| L2: pool with 25% more seats    |       | −95%        | −63%        | −79%        | +57%         | +116%   | +211%        |
| L3: caste quotas, income filter |       | −97%        | +32%        | +37%        | 0%           | 0%      | −1%          |

## D Stylized specification: rank probabilities

**Table 18:** Lever effects and rank probabilities under priors over effect sizes, stylized specification (200 runs).

| Lever | Change | 95% CI      | 1   | 2   | 3   | 4   | 5   | 6   | 7   |
|-------|--------|-------------|-----|-----|-----|-----|-----|-----|-----|
| L1    | −69%   | −72 to −67% | 4%  | 35% | 42% | 18% | 0%  | 0%  | 0%  |
| L2    | −37%   | −40 to −35% | 0%  | 0%  | 3%  | 30% | 50% | 18% | 0%  |
| L3    | −88%   | −90 to −87% | 82% | 14% | 3%  | 0%  | 0%  | 0%  | 0%  |
| L4    | −31%   | −34 to −28% | 0%  | 0%  | 2%  | 13% | 32% | 43% | 10% |
| L5    | −73%   | −76 to −69% | 12% | 44% | 27% | 10% | 4%  | 4%  | 0%  |
| L6    | −18%   | −21 to −16% | 0%  | 0%  | 0%  | 0%  | 6%  | 28% | 66% |
| L7    | −48%   | −55 to −42% | 1%  | 8%  | 24% | 29% | 8%  | 10% | 21% |

**E Stylized specification: central and high-mobilization regimes****Table 19:** Stylized specification, central regime, 50 paired runs. Columns as in Table 9.

|        | Scenario                                           | Peak | 10th–90th  | Change | 95% CI      | Ever | Deaths |
|--------|----------------------------------------------------|------|------------|--------|-------------|------|--------|
| Base V | Abrupt ₹8L income-only switch                      | 46.1 | 14.9–118.9 | –      | –           | 1.29 | 18.5   |
|        | Validation: symbolic threat only (2018-type shock) | 13.3 | 6.0–37.6   | –70%   | –75 to –65% | 0.47 | 5.5    |
| L1     | Grandfather current cohorts + 10-year glide        | 15.6 | 6.6–41.5   | –66%   | –73 to –62% | 0.50 | 5.5    |
| L2     | Expand seats by a quarter (EWS precedent)          | 24.1 | 9.8–65.8   | –44%   | –50 to –37% | 0.74 | 11     |
| L3     | Keep caste sub-quotas, add income filter           | 2.9  | 1.7–4.3    | –94%   | –96 to –92% | 0.10 | 1      |
| L4     | Sub-classify to favour most-deprived               | 26.9 | 11.6–71.4  | –34%   | –39 to –28% | 0.83 | 13.5   |
| L5     | Data-first commission + cross-party consensus      | 8.3  | 4.1–24.2   | –80%   | –84 to –75% | 0.41 | 4.5    |
| L6     | Compensate above-line losers                       | 34.9 | 13.1–93.9  | –18%   | –22 to –14% | 1.08 | 16.5   |
| L7     | Guarantee untouched protections                    | 18.2 | 9.6–55.2   | –49%   | –55 to –42% | 0.68 | 7.5    |
| B1     | Internet shutdowns                                 | 42.8 | 17.2–73.7  | –21%   | –31 to –2%  | 1.41 | 54.5   |
| B2     | Heavy policing and mass arrests                    | 32.9 | 17.4–73.3  | –26%   | –39 to 0%   | 1.61 | 79.5   |
| C1     | Managed transition package (L1+L2+L5+L6+L7)        | 2.2  | 1.4–3.4    | –95%   | –96 to –93% | 0.12 | 1      |
| C2     | Hybrid design package (L3+L4+L1+L5+L6+L7)          | 0.6  | 0.4–0.9    | –99%   | –99 to –98% | 0.04 | 0      |

**Table 20:** Stylized specification, high-mobilization regime ( $\bar{\theta}$  lowered by 0.5), 50 paired runs. Peak and interval in lakh; change is the median paired change with 95% bootstrap interval; cumulative in crore.

|      | Scenario                                           | Peak  | 10th–90th  | Change | 95% CI      | Ever | Deaths |
|------|----------------------------------------------------|-------|------------|--------|-------------|------|--------|
| Base | Abrupt ₹8L income-only switch                      | 209.4 | 92.6–388.4 | –      | –           | 4.14 | 79     |
| V    | Validation: symbolic threat only (2018-type shock) | 66.1  | 18.0–152.5 | –65%   | –73 to –62% | 1.84 | 27     |
| L1   | Grandfather current cohorts + 10-year glide        | 74.2  | 20.0–160.7 | –62%   | –68 to –59% | 1.98 | 30.5   |
| L2   | Expand seats by a quarter (EWS precedent)          | 117.0 | 35.5–258.0 | –46%   | –49 to –41% | 2.50 | 45     |
| L3   | Keep caste sub-quotas, add income filter           | 7.7   | 5.1–12.5   | –96%   | –97 to –96% | 0.29 | 3      |
| L4   | Sub-classify to favour most-deprived               | 106.5 | 40.0–225.0 | –44%   | –48 to –40% | 2.54 | 46.5   |
| L5   | Data-first commission + cross-party consensus      | 42.2  | 13.5–106.7 | –78%   | –80 to –74% | 1.55 | 26.5   |
| L6   | Compensate above-line losers                       | 167.0 | 49.3–327.2 | –15%   | –19 to –13% | 3.80 | 65     |
| L7   | Guarantee untouched protections                    | 110.8 | 34.9–247.2 | –47%   | –52 to –42% | 2.52 | 43     |
| B1   | Internet shutdowns                                 | 110.6 | 55.9–184.4 | –47%   | –50 to –42% | 3.43 | 152.5  |
| B2   | Heavy policing and mass arrests                    | 124.3 | 52.4–245.6 | –37%   | –41 to –34% | 4.20 | 254.5  |
| C1   | Managed transition package (L1+L2+L5+L6+L7)        | 6.4   | 3.6–10.0   | –97%   | –97 to –96% | 0.36 | 4      |
| C2   | Hybrid design package (L3+L4+L1+L5+L6+L7)          | 1.9   | 1.3–2.9    | –99%   | –99 to –99% | 0.12 | 1      |

## F Code map

| File                                           | Contents                                                                                              |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| merit_allocation/                              | Rank-list data, latent merit fit, over-and-above and reserved-first choice rules, merged-pool regimes |
| data_pipelines/                                | Rank-list extraction, GDELT download and filtering, episode definitions, calibration targets          |
| protest_simulation/synthetic_population.py     | National population, tiers, income line, eligibility flags, neighbourhoods                            |
| protest_simulation/geography.py                | District population from Census 2011 and NFHS-5                                                       |
| protest_simulation/model_parameters.py         | Parameters and structural options                                                                     |
| protest_simulation/protest_campaign.py         | Daily dynamics                                                                                        |
| protest_simulation/policy_interventions.py     | Scenario catalogue at reference values                                                                |
| protest_simulation/intervention_priors.py      | Priors over intervention effects                                                                      |
| protest_simulation/episode_simulation.py       | Historical episodes as model shocks                                                                   |
| protest_simulation/grounded_specification.py   | Grounded specification                                                                                |
| protest_simulation/monte_carlo.py, parallel.py | Paired runs, bootstrap intervals, multiprocessing                                                     |
| experiments/                                   | One script per analysis in this paper                                                                 |
| tests/                                         | Regression, property and reimplementations tests                                                      |